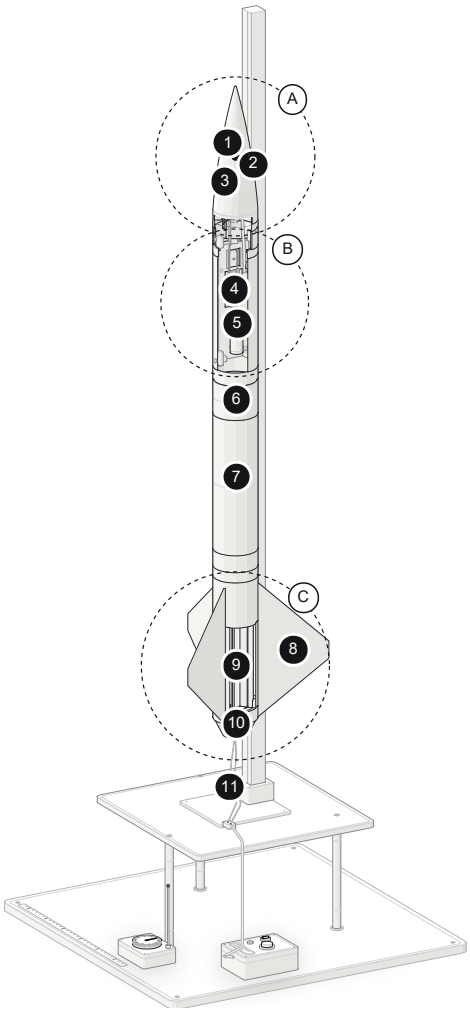
Helios



How does a small rocket fly, and bring itself back?

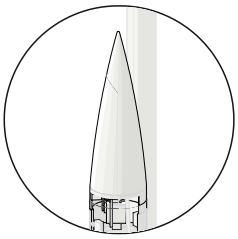
A 38 mm motor burns two grains of potassium nitrate and sorbitol for 1.3 s and throws the 1.15 kg rocket off a 1.2 m rail. It coasts to about 800 m. There a small computer in the main body sees the air pressure start to rise, turns a servo in the nose, and two springs kick a side door open so the parachute can come out.

218 N·s TOTAL IMPULSE	170 N PEAK THRUST
800 m APOGEE	1.15 kg AT LIFTOFF
6.5 m/s UNDER THE CANOPY	

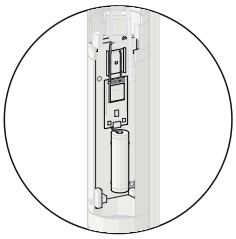
PARTS

01	Side-door nose	Holds the servo latch, the springs and the packed parachute.
02	Door	One quarter of the nose; its foot is held by the servo's arm.
03	Parachute	Packed in a bag on a ramp; slides out once the door is gone.
04	Avionics bay	ESP32, BMP280 barometer and an 18650 cell on a sliding deck.
05	Main body	Anchors the Kevlar shock cord and carries the bay.
06	Coupler sleeve	Joins the main body to the lower body on their spigots.
07	Lower body	The middle tube, between the coupler and the fin can.
08	Fin can	Three fins through the wall, around the motor tube.
09	38 mm KNSB motor	Two BATES grains burning from the core and the ends.
10	Boattail motor cap	Holds the motor in and tapers the tail.
11	Launch pad and rail	A 1010 rail, a blast plate, the controller and ground station.

CLOSE-UPS



(A) **The latch**
A servo arm stands in a groove under the door's floor block. Two springs wait, squeezed to 11.4 mm, behind it.

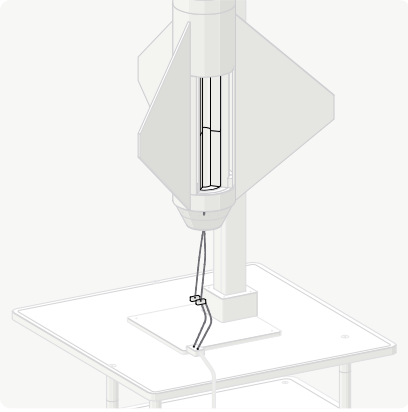


(B) **The avionics**
The ESP32 and the BMP280 ride on a deck over the 18650 cell.



(C) **The motor**
Two BATES grains in a 38 mm case, a nozzle with a 7.5 mm throat.

How it works



01

On the rail

The rocket hangs on its two guides on a 1.2 m rail over a blast plate. The igniter leads run from the nozzle to the controller's clips.



02

The grains burn

Both grains burn from their cores and ends at about 2.8 MPa. The motor gives up to 170 N for 1.3 s and the rocket leaves the rail at 16.6 m/s.



03

The barometer watches

The BMP280 in the bay reads the pressure many times a second. When the altitude has dropped for three readings in a row, the ESP32 knows it is past the top.



04

The door kicks off

The servo turns 60° and its arm leaves the door's groove. Two springs push the door off and it swings away, and the parachute bag slides out.



05

Under the canopy

The Kevlar cord, anchored in the main body, catches the canopy. The rocket comes down at about 6.5 m/s.

WHY A SERVO DOOR?

Most small rockets push their nose off with a powder charge timed from the motor. Helios's ESP32 waits until its barometer shows the rocket has started to fall, then turns a servo, so the parachute opens at the top whatever the motor did, with nothing to burn.

SPECIFICATION

Airframe	65.5 mm, five printed sections	Apogee	about 800 m at T+12 s
Motor	38 mm KNSB, 2 BATES grains	Recovery	servo side door, 0.76 m canopy
Burn	1.3 s, 218 N·s, 170 N peak	Descent	6.5 m/s
Rail exit	16.6 m/s off 1.2 m		